

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2

**ST. ANDREW'S JUNIOR COLLEGE
2017 JC2 PRELIMS**

H2 BIOLOGY**9744/3****Paper 3 (Mark Scheme)**

Monday

18th September 2017

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

For Examiners' Use	
Section A	
1	/17
2	/18
3	/15
Total	/50

Conceptual error (CE)	Lack of Keywords (K)	Misreading the question (Q)

This document consists of **14** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

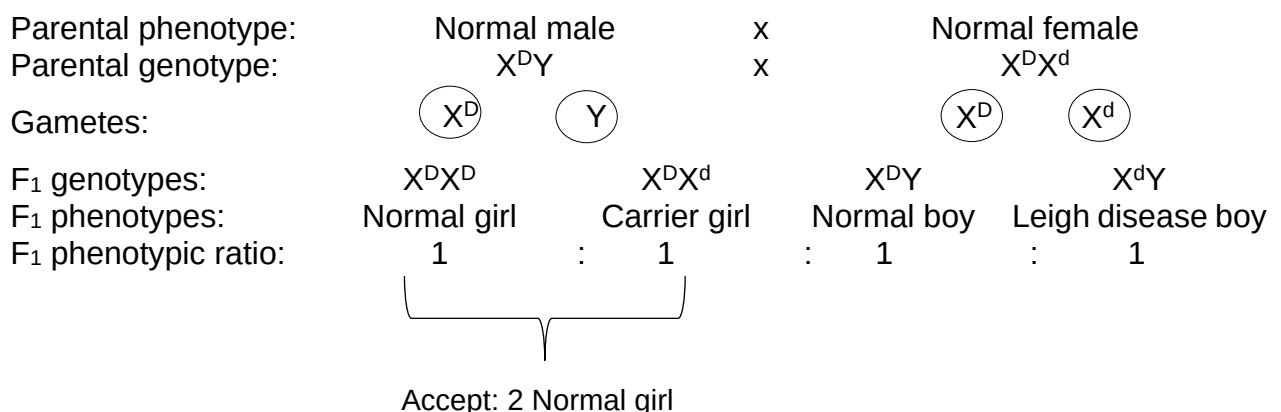
QUESTION 1

Leigh disease is an inherited neuro-metabolic disorder that affects the central nervous system. As the disease progresses, the muscular system is debilitated throughout the body, as the brain cannot control the contraction of muscles. Leigh disease can be caused by a deficiency of the pyruvate dehydrogenase complex (PDHC), most commonly due to a mutation in the X-linked gene, PDHA1, which codes for the PDHC α -subunit.

A married couple (both of whom are normal individuals) is concerned about their chances of having a child with Leigh disease because the woman's father had the disease.

- (a) Draw a genetic diagram with appropriate symbols to determine the probability that they will have an affected child.

.....[6]



Probability of having an affected child = $\frac{1}{4}$

Mark scheme:

- 1 Parental genotypes (showing X-linked symbols)
- 2 Gametes (circled)
- 3 F₁ genotypes
- 4 F₁ phenotypes corresponding to genotypes
- 5 F₁ Phenotypic ratio
- 6 Probability = $\frac{1}{4}$ [Award mark if genotype is wrongly written]

[No marks for Points 1-5 if genotype is written wrongly e.g. "X" represents normal allele; "x" represents diseased allele; "X" represents allele for disease; "X⁰" represents allele for no disease]

200 couples with the same genotypes as the above-mentioned couple were included in a genetic study of the Leigh disease. They had a total of ~~326~~ **322** children with the following phenotypes:

85 normal girls
 70 carrier girls
 89 normal boys
 78 boys with Leigh disease

(b) A chi-squared test was carried out to test the significance of the difference between the observed (O) and the expected (E) results.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

(i) Calculate the χ^2 value. Show your working.

.....[2]

$$\begin{aligned}\chi^2 &= \frac{3.5^2}{80.5} + \frac{11.5^2}{80.5} + \frac{7.5^2}{80.5} + \frac{3.5^2}{80.5} \\ &= 2.596 \text{ (3 d.p.) (Follow the chi sq table) / 2.60 (2 d.p.)}\end{aligned}$$

OR

$$\begin{aligned}\chi^2 &= \frac{(85-80.5)^2}{80.5} + \frac{(70-80.5)^2}{80.5} + \frac{(89-80.5)^2}{80.5} + \frac{(78-80.5)^2}{80.5} \\ &= 2.596 \text{ (3 d.p.) (Follow the chi sq table) / 2.60 (2 d.p.)}\end{aligned}$$

- 1 Correct working
- 2 Answer given to 3 d.p. or 2 d.p.

Fig. 1

Degrees of Freedom	probability								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28

(ii) Fig. 1 shows the table of probabilities. State the probability that the observed results does not differ significantly from the expected results.

.....[1]

- 1 $0.50 > p > 0.25$

(iii) State what conclusions may be drawn from the probability found in **(ii)**.

.....[3]

[No error carried forward if (ii) is wrong]

- 1 Since $0.50 > p > 0.25$ is greater than 0.05,
- 2 there is no significant difference between the observed and the expected **values** [Reject: ratio] at the 5% level.
Any observed differences is due to chance.
- 3 The predicted ratio of 1 : 1 : 1 : 1 is correct / the inheritance is sex-linked
[Reject: there is mendelian inheritance]

[Reject: there is independent assortment of the 2 genes → this is a monohybrid cross and independent assortment is not relevant to this case]

Leigh disease can also be due to a mutation in the mtDNA (mitochondrial DNA) affecting the MT-ATP6 gene. The mutated allele results in the production of a non-functional subunit in the ATP synthase complex which allows protons to pass through the channel without generating the proton motive force.

- (c) State whether a mitochondrion containing the mutated allele would have a higher, lower or equal oxygen consumption rate relative to a mitochondrion which has the normal allele.

.....[1]
1 Equal / Higher

- (d) Explain your answer in (c).

.....[2]

[Explanation for Equal rate]

1. Oxygen acts as a final electron acceptor [Reject: final electron carrier] (in oxidative phosphorylation).
2. Since the **flow of electron through the ETC is not affected** by the mutation / the **donation of electrons from NADH and FADH₂ to the ETC is not affected**, (the rate of oxygen consumption in the mutated mitochondrion will remain the same.)

OR

[Explanation for Higher rate]

1. Ref. reduction in ATP production in oxidative phosphorylation,
2. Therefore, rate of oxidative phosphorylation increases to **regenerate NAD⁺ for use in glycolysis** for alternative ATP production

The DNA sequence of the MT-ATP6 gene has remained relatively conserved over a long evolutionary period. It is present in a wide diversity of organisms, ranging from the simplest worms to the great apes. Differences in the nucleotide sequence of the gene in different species can be identified using multiple sequence alignments.

- (e) Discuss how the comparison of MT-ATP6 DNA nucleotide sequences of different species demonstrates evolutionary change.

.....[2]

- 1 The **similarities** in the nucleotide sequences shows molecular homology / which suggests inheritance from a common ancestor.
- 2 As the descendants of same species evolve independently, more and more differences (mutations) are **accumulated** in their DNA.
- 3 Resulting in divergent evolution and descent with modification.

Examiner's comment:

Students who stated that “the more similar the sequences are, the more closely related the organisms are to each other” have misinterpreted the question to mean “explain how molecular homology demonstrates evolutionary relationship between different species”

[Q1: 17 marks]

QUESTION 2

Celiac disease is an autoimmune disorder that is caused by an improper immune response to the protein gluten, found in wheat, rye, and barley, that damages the lining of the small intestine. There is no cure for celiac, and the only effective treatment is a gluten-free diet.

Recent studies indicated that harmless intestinal viruses, such as the reovirus, can cause the immune system to overreact to gluten, raising the possibility of such viruses contributing to the development of the disease.

Fig. 2.1 shows the general structure of a reovirus. Unlike Human Immuno-deficiency Viruses (HIV), reoviruses are not retroviruses.

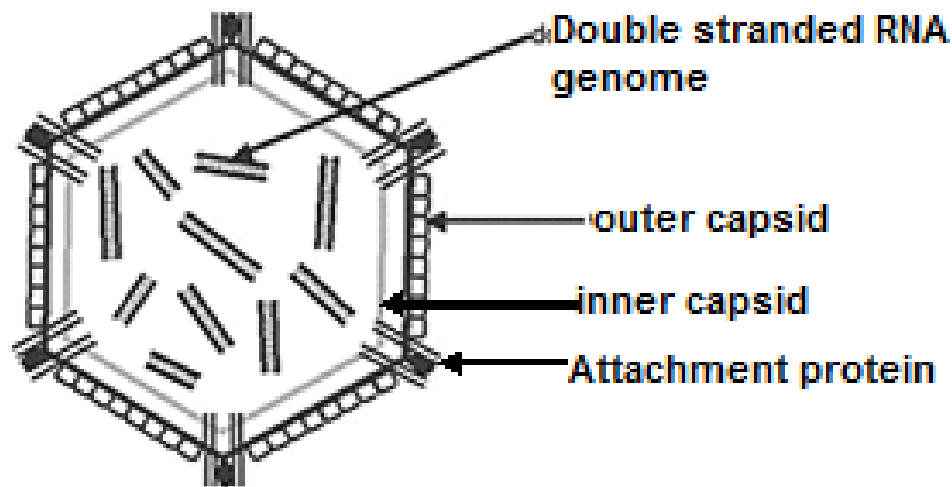


Fig. 2.1

(a) With reference to Fig. 2.1, contrast the structure of the reovirus with that of HIV.[2]

- 1 Reovirus has no viral envelope but HIV has a viral envelope
- 2 Reovirus has double-stranded RNA genome but influenza virus has single-stranded RNA genome
- 3 Reovirus has 2 capsids while HIV has 1 capsid
- 4 Segmented genome vs HIV non-segmented genome
- 5 Absence of viral enzymes in Reovirus but presence of viral enzymes (list at least 2: integrase, protease, reverse transcriptase) in HIV

[Any 2]

(b) Suggest how new double-stranded viral RNA genome is synthesized during the reproductive cycle of a reovirus.[2]

- 1 Viral RNA **unzips** to form single stranded RNA
- 2 **Viral** RNA-dependent **RNA polymerase** is used to make new RNA strands from RNA template by **complementary base pairing**

Researchers also looked at patients with celiac disease and found that they had much higher levels of antibodies against reoviruses than those without the disease.

Those with higher levels of antibodies also had higher levels of the molecule IRF-1 (interferon regulatory factor 1), a regulator of gene transcription which plays a key role in the loss of gluten tolerance.

(c) Describe three ways in which the structure of antibodies contribute to its function.[3]

- 1 **Constant** region; determines the **class** of antibody (e.g. IgG, IgA, IgM etc) / allows binding to phagocyte (e.g. macrophage/neutrophil)
- 2 **Variable** regions (of each light chain and heavy chain); forms an antigen binding site that provides a lock-and-key fit for **specific binding** to a particular epitope of an antigen.
- 3 **Disulfide bridges** between chains; **links** the heavy and light chains together
- 4 **Hinge region**; provides **flexibility** to change orientation/movement to bind antigen

Examiner's comment:

Students are reminded to write both structure and function.

[Reject: different combinations of heavy and light chains allow binding to different antigens on pathogens → this addresses why a huge variety of pathogens can be recognised]

Fig. 2.2 shows the structure of an immunoglobulin gene which codes for antibodies.

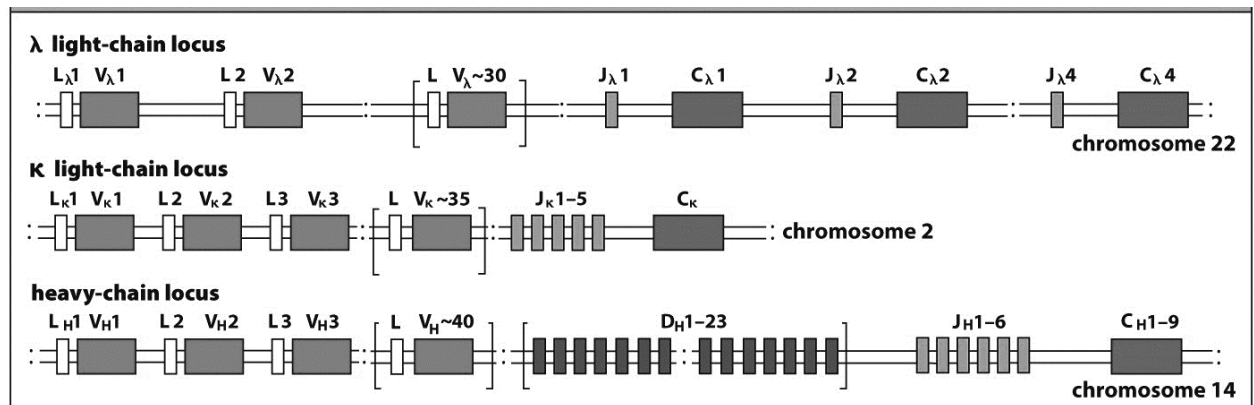


Fig. 2.2

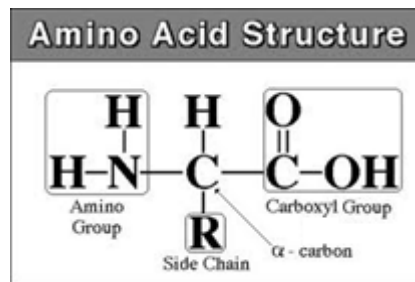
(d) With reference to Fig. 2.2, explain how variability at the DNA level result in variability in the antigen-binding sites of antibodies.

-[3]
- 1 **Somatic recombination** (at the DNA level) of **both light and heavy chain genes** occurred
 - 2 only **one V segment** and **one J segment** are arranged together, along with the C segment for the **light chain gene**
 - 3 only **one D segment** and **one J segment** are first arranged together, followed by **one V segment** being arranged next to the pre-arranged DJ region, along with the C segments for the **heavy chain gene**

Reject: class switching as it does not affect the variable regions

(e) Antibodies are proteins. Draw a diagram of the monomer which makes up antibodies. There is no need to annotate the diagram.

-[1]
- 1 Correct structure of amino acid drawn showing the amino group, carboxyl group, alpha carbon, R group side chain and hydrogen atom



[Accept: zwitterion diagram]

[Reject: repeating n units of monomers linked together $(-\text{HN}-\text{C}-\text{COO}-)_n$]

Antibodies are secreted by plasma cells.

Fig. 2.3 shows a plasma cell and a B lymphocyte.

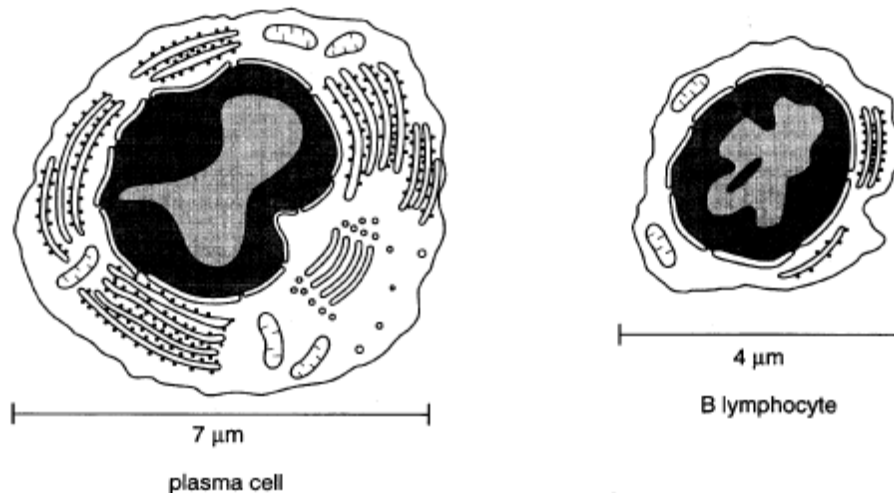


Fig. 2.3

(f) With reference to Fig. 2.3,

(i) State 2 ways in which the structure of plasma cell differs from the B lymphocyte.[2]

1. plasma cell has **Golgi apparatus** but B lymphocyte does not;
/ plasma cell has Golgi **vesicles** / secretory, vesicles but B lymphocyte does not;
2. plasma cell has **more** (rough) endoplasmic reticulum;
3. plasma cell has **more** ribosomes;
4. plasma cell has **more** mitochondria;
5. plasma cell is larger at 7µm vs 4µm / contains **more** cytoplasm;
[Any 2]

(ii) Explain the reasons for the differences you described.[2]

check answer against **(i)**

1. [plasma cell has **Golgi apparatus** but B lymphocyte does not] Enables packaging / **chemical** modification/ adding carbohydrate / making glycoprotein to form antibody
/ [plasma cell has Golgi **vesicles** / secretory, vesicles but B lymphocyte does not] Enables for transport of antibody to plasma membrane for secretion from cell
2. [plasma cell has **more** (rough) endoplasmic reticulum] Enables **fast** / **large** production of protein / antibodies
3. [plasma cell has **more** ribosomes] Enables **fast** / **large** production of antibodies which are **proteins**
4. [plasma cell has **more** mitochondria] Provides **more ATP** which releases energy upon hydrolysis for production of protein / antibody
5. [plasma cell is **larger** / contains **more** cytoplasm] Allows **more space** for organelles e.g mitochondria, rER, Golgi apparatus etc

[Any 2 corresponding to **(i)** responses]

A student researcher tried to reproduce the results of the study regarding elevated antibody levels against reoviruses. She studied the levels of antibodies in 3 human subjects infected with reoviruses and another 3 who are not infected.

Table 2.4 shows his results.

Table 2.4

	Antibody titre / unit ml ⁻¹
Subjects not infected with reovirus	30 ± 10
Subjects infected with reovirus	122 ± 25

(g) With reference to the results on subjects not infected with reoviruses, explain what is standard deviation and its implications.

-[3]
- 1 Standard deviation is the **deviation from the mean** antibody titre
 - 2 and determines the **range** of antibody titre observed in subjects not infected with reovirus
 - 3 [Evidence]The antibody titre ranges from 20 ($30 - 10$) to 40 ($30 + 10$) unit ml^{-1} in subjects not infected with reovirus / the antibody titre deviates 10 unit ml^{-1} from the mean of 30 unit ml^{-1} ;
 - 4 Standard deviation is an indication of the reproducibility of the data collected. Since standard deviation for subjects not infected is high, results are not very reproducible / the data don't change too much when the experiment is **repeated**

[Q2: 23 18 marks]

QUESTION 3

- (a) A student investigated growth in the roots of broad bean, *Vicia faba*. The student cut sections of the root tip of this plant and viewed them with a light microscope.

Fig. 3.1 is a photomicrograph of one of the sections. The cell labelled **D** is in interphase.

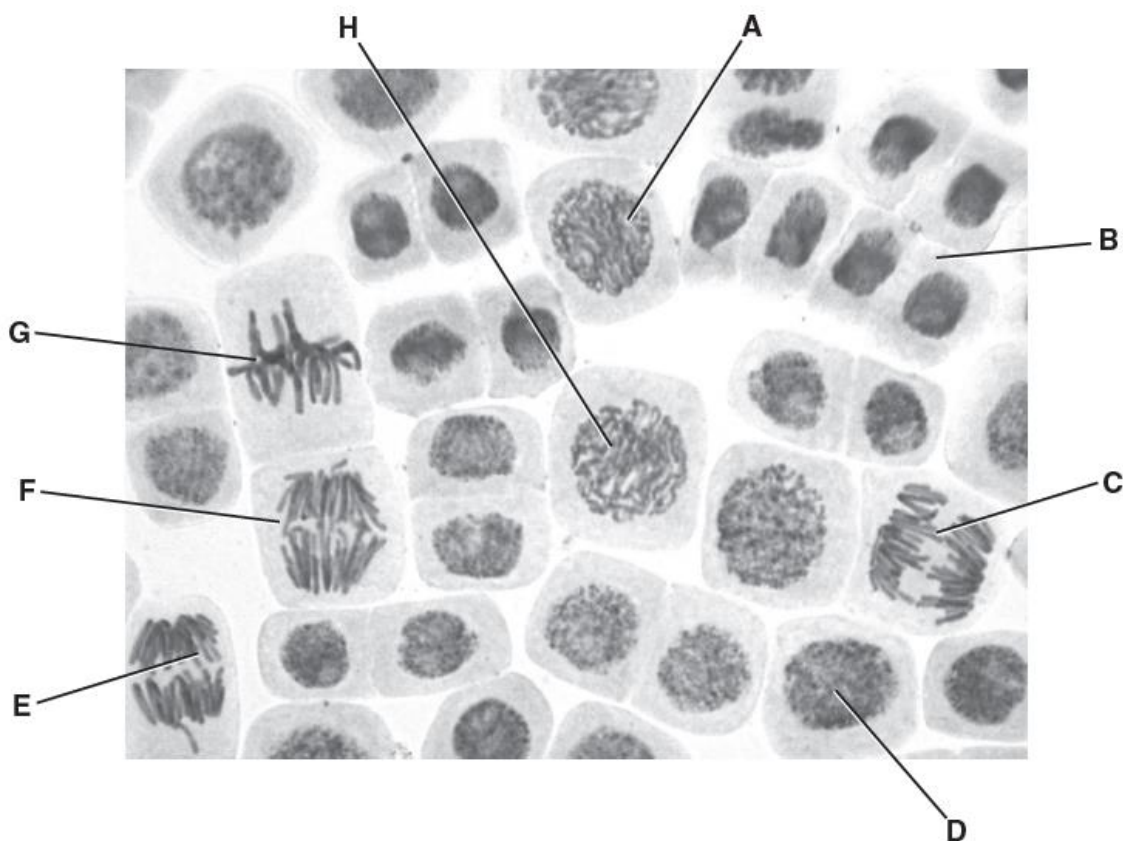


Fig. 3.1

Complete the table below by:

- naming the stages of mitosis in the correct sequence following interphase
- identifying **one** example from the cells labelled **A** to **H** that is in each stage of mitosis that you have named.

[3]

Mark scheme:

- 1 one mark for the stages of the cell cycle in the correct sequence ie. Prophase, metaphase, anaphase and telophase under column heading “stage of mitosis” ;
- 2 one mark for each two correct matching of each stage with a cell ; (mark for 1st answer given)

stage of mitosis	label from Fig. 3.1
prophase	A / H ;
metaphase	G ;
anaphase	C / E / F ;
telophase	B ;

- (b)** In animal cells, centrioles are responsible for assembling microtubules to make the spindle at the beginning of mitosis.

Describe the role of the spindle during mitosis.

-[3]
- 1 (kinetochore) microtubules attached to kinetochore on centromere of duplicated chromosomes (during prophase) ;
 - 2 arranging / aligning / orienting / AW, duplicated chromosomes at the equator / metaphase plate ; **[REJECT: centre of the cell]**
 - 3 centromere divides ; microtubules **shorten** ;
sister chromatids/chromosomes pulled to opposite poles [Reject: opposite ends] (of cell) ;
 - 4 Ref. Equivalent and complete collection of chromosomes at both poles of the cell;

OVP:

- 5 Overlapping/polar microtubules (slide past each other) elongate the cell;

- (c)** Scientists investigated three genes, **P**, **Q** and **R**, involved in controlling cell division. They studied the effect of mutations in these genes on the risk of developing lung cancer.

- (i)** Suggest the differences in the cell cycle of a cancer cell compared with that of a normal cell of the same type.

-[2]
- 1 Cell cycle shorter / interphase shorter / division more frequent ; [Reject: cell cycle is irregular / inappropriate]
 - 2 (cell cycle) **checkpoints** not controlled / unregulated ;
/ uncontrolled cell division / AW ;
 - 3 Cell cannot be stimulated to undergo apoptosis (even when errors occur during the cell cycle);

The scientists analysed genes, **P**, **Q** and **R** from healthy people and people with lung cancer.

- If a person had at least one copy of normal allele for a gene, they used the symbol N.
- If a person had two mutant alleles for a gene, they used the symbol M.

They used their data to calculate the risk of developing lung cancer for people with different N and M genotypes. A risk value of 1.00 indicates no increased risk. **Table 3.1** shows the scientists' results.

Table 3.1

Gene P	Gene Q	Gene R	Risk value of developing lung cancer / AU
N	N	N	1.00
M	N	N	1.30
N	N	M	1.78
N	M	N	1.45

N = at least one copy of the normal allele is present

M = two copies of the mutant allele are present

(ii) What do these data suggest about the relative importance of the mutant alleles of genes **P**, **Q** and **R** on **increasing** the risk of developing lung cancer?

.....[2]

1 [general trend] order of mutant alleles that increases the risk of developing lung cancer from highest to lowest : **R > Q > P** ;

2 [quote data to support ans] Being **homozygous** for the mutation in **R** produces **highest** risk of 1.78 A.U.; in **Q** produces **next highest** risk of 1.45 A.U.; in **P** produces **least** risk of 1.30 A.U.

Chemotherapy is the use of a drug to treat cancer. An experiment was set up to study a new drug, SA128, which kills dividing cells. A group of four men suffering from lung cancer was given the drug. The number of cancer cells per unit volume of blood was measured right before treatment and again after the 2-week treatment.

Table 3.2 : Effect of SA128 on cancer cells.

Subject	Number of cancer cells per unit volume of blood					
	TSH	OON	CCK	TTS	Average	Standard deviation
Before drug treatment	2000	1600	1800	1200		
After drug treatment	1000	500	800	300		

(iii) Calculate the **average** and complete Table 3.2.

[1]

Subject	Number of cells per unit volume of blood					
	TSH	OON	CCK	TTS	Average	Standard deviation
Before drug treatment	2000	1600	1800	1200	1650	342
After drug treatment	1000	500	800	300	650	311

- (iv) Using the formula below, calculate the **standard deviations** of number of cancer cells per unit volume of blood before and after drug treatment, SA128. Complete Table 3.2. Express your answers to **3 significant figures**.

.....[1]

standard deviation $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$

Legend

$\sum$ is 'Sum of'

x is observation

$\bar{x}$ is the mean

n is the sample size (number of observations)

- (v) Using the formula below as well as the average and standard deviation values calculated in (c)(iii) and (c)(iv),

$$t = \frac{(X_1 - X_2)}{\sqrt{\frac{(S_1)^2}{n_1} + \frac{(S_2)^2}{n_2}}}$$

Where:

- x_1 is the mean of sample 1
- s_1 is the standard deviation of sample 1
- n_1 is the sample size of sample 1
- x_2 is the mean of sample 2
- s_2 is the standard deviation of sample 2
- n_2 is the sample size in sample 2

Calculate the $t_{\text{calculated}}$ value.

.....[1]
1. One mark for correct calculation of $t_{\text{calculated}}$ value (step 2) – 4.327 (3 dp)

Examiner's comments:

No error carried forward if values calculated in (iii) and (iv) are wrong unless wrong dp.

- (vi) Using the critical t-values in **Table 3.3**, determine if there is a significant difference in the results using the new drug, SA128.

.....[2]
1 One mark for correct calculation of **DoF** (Ans: **6**) and identification of t_{critical} value of **1.943** (step 3, 4)
2 One mark for correct comparison of $t_{\text{calculated}}$ and t_{critical} and making a correct conclusion that there is significant difference in the results for before and after treatment with drug, SA128. (step 5)

$t_{\text{calculated}}$ of 4.327 is higher than t_{critical} (1.943) . There is a significant difference between the results before and after treatment with drug, SA128.

Examiner's comments:

No error carried forward if DF or T critical is wrong

Working for Reference:

Step 1: Null hypothesis:

There is no significant difference between the results before and after treatment using the new drug, SA128.

Step 2: Calculation of experimental t-value

$$x_1 - x_2 = 1650 - 650 = 1000$$

$$(s_1)^2 = (342)^2 = 116964$$

$$(s_1)^2 / n = 116964 / 4 = 29241$$

$$(s_2)^2 = (311)^2 = 96721$$

$$(s_2)^2 / n = 96721 / 4 = 24180$$

$$t = 1000 / \sqrt{29241 + 24180}$$

$t_{\text{calculated}} = 4.327$ (3 dp according to T table)

Step 3: Calculation of Degrees of Freedom

Degrees of Freedom is sum of sample sizes for control and test minus 2 = $8 - 2 = 6$.

Step 4: Stating of t_{critical} value (from Table of significance, looking at $p=0.05$)

t_{critical} value for 6 DoF and p value 0.05 is **1.943**.

Step 5: Comparison between $t_{\text{calculated}}$ and t_{critical}

$t_{\text{calculated}}$ of 4.327 is higher than t_{critical} (1.943) . Null hypothesis is not accepted.

Conclusion:

There is a significant difference between the results before and after treatment with drug, SA128.

Table 3.3: t-test table of t critical values

df	.10	.05
1	3.078	6.314
2	1.886	2.920
3	1.638	2.353
4	1.533	2.132
5	1.476	2.015
6	1.440	1.943
7	1.415	1.895
8	1.397	1.860
9	1.383	1.833
10	1.372	1.812
11	1.363	1.796
12	1.356	1.782
13	1.350	1.771
14	1.345	1.761
15	1.341	1.753
16	1.337	1.746
17	1.333	1.740
18	1.330	1.734
19	1.328	1.729
20	1.325	1.725
21	1.323	1.721
22	1.321	1.717
23	1.319	1.714
24	1.318	1.711
25	1.316	1.708
26	1.315	1.706
27	1.314	1.703
28	1.313	1.701
29	1.311	1.699
30	1.310	1.697
40	1.303	1.684
60	1.296	1.671
120	1.289	1.658
c	1.282	1.645

[Q3 Total: 15]

Section B

Answer **one** question in this section.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 4 (a)** The effects of global warming on the spread of malaria beyond the tropics is a debatable issue. Discuss the arguments/evidences that support your stance on the matter, and provide a balanced account of counter-arguments. [12]

- (b)** Discuss the effects of climate change (as a result of greenhouse gas emissions. [13]

[Q4 Total: 25]

- 5 (a)** Explain what is meant by primary, secondary, tertiary and quaternary structure of haemoglobin. [13]

- (b)** Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [12]

[Total: 25]

4(a) The effects of global warming on the spread of malaria beyond the tropics is a contentious issue. Discuss the arguments/evidences that support your stance on the matter, and provide a balanced account of counter-arguments. [12]

Arguments and evidences **supporting** the theory that global warming causes spread of malaria beyond the tropics

- 1 Malaria can only occur in climates where mosquitoes are present to transmit the disease.
- 2 Global warming leads to temperatures in the sub-tropical regions becoming more optimal for both mosquitoes and malaria parasite.
- 3 For the optimal development of Anopheles mosquitoes, the temperatures should be **around 20 to 30°C**.
- 4 Ref. higher temperature leads to higher rate of **development** / metabolic rate / survival of **both mosquitoes**
- 5 Ref. **parasite** plasmodium development peaks at 26°C (thus, the closer the sub-tropical temperatures is to 26°C, the higher the rate of plasmodium development)
- 6 In addition, global warming can lead to higher humidity (in the sub-tropics), more stagnant pools of water available for mosquitoes to lay eggs
- 7 [Observation on sub-tropical regions] There was an apparent association between an **increased incidence in malaria and increased minimum temperatures** during December - January of the preceding year.
- 8 [Observation on sub-tropical regions] A rise in minimum temperature of 1°C, from 15.5 to 16.5°C could account for 24 additional cases per 1000 of population.
- 9 [concluding remark] The above effects of temperature and climate on malaria have led many to believe that **global warming will result in the spread of this disease**, to higher **altitudes** and higher **latitudes** (sub-tropical regions) where it is cooler.
- 10 AVP

[Max 8 marks]

Arguments and evidences **against** the theory that global warming causes spread of malaria beyond the tropics

- 1 (There is common assumption that a faster development of plasmodium parasite leads to higher rate of transmission. But a latest study shows that temperature has a more complex effect.) As temperature rises, **parasites** do develop faster, but **fewer of them become infectious**.
- 2 In sub-tropical regions, an increase of temperature to above 24°C led to higher parasite development but a decrease in malaria risk.
- 3 This is because the **parasite may not be able to cope** with the **higher temperatures** or
- 4 **Mosquito immune systems may work better** at warmer temperatures.
- 5 Many other arguments and evidence have not taken into consideration other more significant factors in the recent upsurge of malaria, such as the **resistance to**

drugs, poor vector control, changes in land uses and human population growth and migration.

6 Ref. elaboration based on any of the other significant factors

7 AVP:

- a. [Poor vector control] Ref. less use of DDT insecticides → more proliferation of mosquitoes
- b. [Changes in land use] Ref. extension of villages of towns, deforestations, building of dams, human populations have come closer to the wilderness and hence are in closer proximity to mosquito breeding grounds
- c. [Population growth] An increase in population will result in increase in malaria, if there is no simultaneous improvement in healthcare facilities and living standards
- d. [Migration] Migration and human travel can cause the spread of malaria from one area to another (where the people have low immunity to the disease)

QwC: at least 3 points in each set of arguments

4(b) Discuss the effects of climate change (as a result of greenhouse gas emissions).

[13]

- 1 Emission of greenhouse gases result in global warming.
- 2 Which causes the melting of ice sheets and sea ice.

[Any 2]

Rise in sea level

- 3 Melting of ice sheets results in extra water entering the ocean, leading to **increase in sea level.**
- 4 Melting of sea ice can lead to a positive feedback loop which **further accelerates the melting of polar ice caps.**
- 5 As oceans become warmer, the **water also expands, increasing the volume of the ocean water and leading to further rise in sea level.**
- 6 **Low-lying islands / coastal** regions may become **submerged.**

[Any 1 pair]

Stress on fresh-water supplies

- 7 Many island nations will have their supplies of **drinking water reduced** because sea water will invade their freshwater aquifers.
- 8 **Melting of** fresh water (i.e. **ice sheets**) into the sea also **turns the already scarce fresh water into salt water**, decreasing the freshwater availability.
- 9 Global warming also results in **more evaporation**, which leads to increased amount of water in the atmosphere, and hence **heavier rainfall.**
- 10 Which leads to more **rapid movement of water from the atmosphere back to the oceans, reducing our ability to store** and use it.
- 11 At higher temperature, **more precipitation** will occur **as rain** rather than snow.
- 12 **Reservoirs fill quickly** to maximum capacity. **Any excess rain water is lost** as runoff which cannot be stored / Less water will penetrate the ground surface, causing reduced soil moisture and reduced groundwater replenishment.

[Any 2]

Heat-waves and heavy rains

- 13 A warmer climate creates an atmosphere that can **collect, retain, and drop more water, changing weather patterns.**
- 14 This causes wet areas to become wetter and dry areas become drier.
/This can lead to **heat waves** and **heavy rain.**
- 15 **Heatwaves** associated with low humidity may result in **wildfire.**
- 16 **Heavy rain** increases the **amount of runoff** into rivers and lakes, washing sediment, nutrients, pollutants, trash, animal waste, and other materials **into water supplies, making them unusable, unsafe, or in need of water treatment.**

[Any 2]

Death of coral reefs

- 17 Heat stress can cause **coral bleaching** / coral polyps expel the zooxanthellae.
- 18 This is because at higher temperatures, **zooxanthellae photosynthesis is disrupted** / zooxanthellae produces more toxic compounds

- 19 If temperatures remain above the bleaching threshold for prolonged periods of time, **corals will eventually die** from starvation and disease.
- 20 **Ocean acidification** affects hard corals as they cannot absorb the calcium carbonate to maintain their skeletons (and the stony skeletons that support corals will dissolve).

[Any 2]

Migration of fishes and insects

- 21 **Rising ocean temperatures** can directly **affect the metabolism, life cycle, and behaviour of marine species**.
- 22 Many **fish species**, especially their young and larvae, have highly specific temperature ranges and will **move to cooler waters to survive**.
- 23 Warmer temperatures may **disrupt the migratory behaviour and timing of several fish species** (for example, by impeding their ability to orient themselves for effective navigation.)
- 24 **Insect distribution and migration** are also greatly influenced by increased temperatures as they are cold-blooded and will **move to habitats with temperatures that are optimal for their growth and reproduction**.
- 25 These insect migrations have **ramifications for many ecosystems**, e.g. insects play important ecological functions such as pollination, vectors of diseases and parasites, killing of other pest insects or are crop pests themselves (name one examples).

[Any 2]

Release of greenhouse gases in frozen organic matter

- 26 Global warming leads to **accelerated melting of permafrost**.
- 27 As permafrost thaws, the **frozen organic matter starts to decay** and is digested by microbes. The digestion **releases carbon dioxide and methane**.

QwC: Obtain at least 1 mark each from 3 different categories of effects of climate change.

5(a) Explain what is meant by primary, secondary, tertiary and quaternary structure of haemoglobin. [13]

Primary structure

- 1 Refers to the type, number and sequence of amino acids in a linear polypeptide chain
- 2 making up each haemoglobin polypeptide (individual α and β subunits)
- 3 ref (each α -chain is) 141 amino acids long and (each β -chain is) 146 amino acids long
- 4 Peptide bond involved in joining all amino acid monomers together

Secondary structure

- 5 Refers to the folding of the polypeptide into **regular structures**
- 6 α -helices / coiling of polypeptide chain into a regular helical conformation.
- 7 Hydrogen bonds between peptide bonds found within the **same** polypeptide chain, **between C=O group** on the peptide bond of one amino acid the **NH group on peptide bond** of another amino acid

Tertiary structure

- 8 the folding of the polypeptide chain into its unique **3-dimensional shape**; / ref. **globular** shape of haemoglobin
- 9 Amino acids far away in primary structure are brought close together (by R group interaction);
- 10 Non-polar/hydrophobic (side chains of) amino acids are buried in the interior; Polar and charged/hydrophilic (side chains of) amino acids are on the surface;
- 11 Bonds involved include hydrophobic interactions, hydrogen bonds and ionic bonds **between R groups of amino acids** in each polypeptide chain (name at least 2)

Quaternary structure

- 12 Refers to the arrangement of the polypeptide subunits within a protein that is made up of more than one polypeptide chain / spatial arrangement of more than one polypeptide chain
- 13 ref. to the **association** of 2α and 2β subunits to form functional haemoglobin molecule
- 14 Bonds involved include hydrophobic interactions, hydrogen bonds and ionic bonds between R groups of amino acids **in the four subunits / between the 4 polypeptide chains** (name at least 2)

Teachers' comments:

It is important to state the definitions of each level of folding and tailor your points to the haemoglobin case study. Note that **disulfide bonds are not present** in haemoglobin.

QwC: Obtain at least 1 mark each from 4 different levels of protein structure

5(b) Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [12]

Hb^SHb^S individuals

- 1 Individual homozygous for Hb^S / genotype Hb^SHb^S suffer from sickle cell anaemia
- 2 the normal haemoglobin (haemoglobin A) in red blood cells is replaced entirely by abnormal haemoglobin (haemoglobin S).
- 3 Sickled red blood cells tend to stick together and obstruct blood flow in capillaries, depriving multiple organs of oxygen, resulting in organ damage / Sickled red blood cells rupture easily, resulting in anaemia and fatigue. May result in early death.
- 4 Therefore, there is a higher potential for the removal of the Hb^S allele from the gene pool.

Hb^AHb^A individuals

- 5 Hb^AHb^A individuals are at greater risk of dying of malaria compared to heterozygotes.
- 6 Therefore, there is a higher potential for the removal of the Hb^A allele from gene pool.

Hb^AHb^S individuals

- 7 Individual who are heterozygous / genotype Hb^AHb^S suffer from sickle cell trait
- 8 produce both normal and abnormal haemoglobin
- 9 Hb^AHb^S individuals are a selective advantage and are able to survive and reproduce in malaria-infected areas.]
- 10 (This condition is known as heterozygote advantage as) it preserves the Hb^S allele in the population
- 11 A mosquito transmits *Plasmodium*, the malaria parasite to humans. The malaria parasite spends part of its life cycle in red blood cells.
- 12 When malaria parasite invade the bloodstream, the red blood cell that contains haemoglobin S is sickled-shaped and is **quickly destroyed by the body, trapping the parasites within the sickle cell and stopping the infection.**
- 13 The **slowdown in blood flow in sufferers** of sickle cell trait also **hampered the parasite's ability to travel** and rapidly infect new cells.

QwC: Mentioning of how Hb^AHb^A and Hb^SHb^S are at selective disadvantage and how heterozygotes are at selective advantage / Points touching on individuals of each of the 3 genotypes

THE END